



UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

OSSD – Y9

Final Semester Project

Full Stack Web Application using FastAPI, Database & Deployment

Objective

The objective of this final semester project is to help students understand:

- Full-stack web application development
- **FastAPI** backend development
- **PostgreSQL** or **MongoDB** database integration
- Frontend development using **HTML and CSS**
- CRUD API implementation
- Server deployment and database hosting
- Project documentation, presentation, and viva preparation

Task Overview

You are required to design, develop, deploy, and present a complete full-stack web application. Students can choose any project idea of their own choice, but the project must include proper documentation, working code, database integration, server deployment, and final viva/evaluation.

Important Deadlines

Phase	Submission / Activity	Deadline
Phase 1	Documentation Submission	6th June 2026
Phase 2	Code Submission + Deployment	19th June 2026
Phase 3	Evaluation and Viva	26th June 2026

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Phase 1: Documentation Submission

In Phase 1, students must submit a project proposal/document explaining what they are going to build. This document should clearly describe the idea, features, users, database design, APIs, frontend pages, and deployment plan.

Documentation must include:

- *Project Title*
- *Problem Statement*
- *Project Objective*
- *Target Users*
- *Main Features*
- *Frontend Pages*
- *Backend API Endpoints*
- *Database Choice: PostgreSQL or MongoDB*
- *Database Tables/Collections*
- *Technology Stack*
- *Deployment Plan*
- *Expected Output*

Minimum Requirements:

- *At least 5 **frontend pages***
- *At least 10 **backend API endpoints***
- *At least 1 **database***
- *Proper CRUD functionality*
- *Clear database schema or collection design*
- *Plan for server deployment and database hosting*

Phase 2: Code Submission

In Phase 2, students must submit the complete working project code. The project must be properly organized and must run without errors.

Frontend Requirements

The frontend must be developed using HTML and CSS. JavaScript can be used if needed for interaction. The frontend must contain at least **5 pages**.

- *Home Page*
- *About Page*
- *Login/Register Page*
- *Dashboard Page*
- *Add/View/Manage Data Page*

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Students can add more pages according to their project requirements.

Backend Requirements

The backend must be developed using **FastAPI**. The backend must contain at least **10 API endpoints**. CRUD APIs are allowed and recommended.

- *Create user / create record*
- *Login user / authenticate user*
- *Add record*
- *View all records*
- *View single record*
- *Update record*
- *Delete record*
- *Search records*
- *Filter records*
- *Dashboard/statistics API*

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Database Requirements

Students must use either **PostgreSQL** or **MongoDB**. The database must store real project data and must be connected with the FastAPI backend.

Supabase Requirement

If students are using PostgreSQL, they are encouraged to host their database on **Supabase**.

Students should:

- *Create a Supabase project*
- *Create database tables*
- *Connect FastAPI with Supabase PostgreSQL*
- *Perform CRUD operations through FastAPI APIs*
- *Use environment variables for database URL and credentials*

MongoDB Option

If students are using MongoDB, they may use MongoDB Atlas for cloud database hosting.

Deployment Requirements

Students are required to deploy their project online. The deployed project should be accessible through a live link.

Backend Deployment Options:

- *Render*
- *Railway*
- *Heroku*
- *AWS*
- *Any other cloud platform*

Frontend Deployment Options:

- *Vercel*
- *Netlify*
- *GitHub Pages*

Database Hosting:

- *PostgreSQL projects should preferably use **Supabase***
- *MongoDB projects may use **MongoDB Atlas***

Deployment Deliverables:

- *Live project link*
- *Backend API link*
- *GitHub repository link*
- *Database screenshots or configuration screenshots*

Suggested Project Ideas

- *Student Management System*
- *Library Management System*
- *Online Book Store*
- *Hospital Appointment System*
- *Task Management System*
- *Expense Tracker*
- *Event Management System*
- *Course Registration System*
- *Restaurant Ordering System*
- *Complaint Management System*
- *Job Portal System*
- *Inventory Management System*
- *Hotel Booking System*
- *Online Quiz System*
- *Fitness Tracker System*

Recommended Folder Structure

Students may follow the folder structure below:

```
FinalProject/  
■  
■■■ frontend/  
■   ■■■ index.html  
■   ■■■ about.html  
■   ■■■ login.html  
■   ■■■ dashboard.html  
■   ■■■ manage.html  
■   ■■■ style.css  
■  
■■■ backend/  
■   ■■■ main.py  
■   ■■■ models.py  
■   ■■■ database.py  
■   ■■■ schemas.py  
■   ■■■ requirements.txt  
■  
■■■ README.md  
■■■ screenshots/
```

Phase 3: Evaluation and Viva

In Phase 3, students will present their project and answer viva questions. Students must be able to explain their own idea, code, APIs, database, and deployment.

Evaluation will be based on:

- *Project idea and understanding*
- *Quality of documentation*
- *Frontend design*
- *Backend API implementation*
- *Database design and integration*
- *CRUD functionality*
- *Deployment success*
- *Code structure*
- *Error handling*
- *Final presentation*
- *Viva performance*

Students must be able to explain:

- *What their project does*
- *Why they selected this project*
- *How frontend connects with backend*
- *How APIs work*

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- *How database stores data*
- *How CRUD operations are performed*
- *How deployment works*
- *How Supabase/MongoDB Atlas is connected*
- *What challenges they faced*

Important Rules

- *Project must be original*
- *Copy-paste projects will not be accepted*
- *Each student/group must explain their own code*
- *APIs must be tested properly*
- *Database must be connected successfully*
- *Project must be deployed online*
- *Frontend should be clean and understandable*
- *Final submission must include code, documentation, screenshots, and deployed links*

Final Submission Requirements

Submit the following:

- *Phase 1 documentation PDF*
- *Complete project source code*
- *GitHub repository link*
- *Live frontend deployment link*
- *Live backend API link*
- *Database screenshots/configuration screenshots*
- *Project screenshots*
- *README file with setup instructions*

Email submissions are not acceptable unless specifically allowed by the instructor.

Minimum Submission Checklist

Before final submission, students must ensure:

- *Documentation is complete*
- *Frontend has at least 5 pages*
- *Backend has at least 10 APIs*
- *Database is connected*
- *CRUD operations are working*
- *Project is deployed online*
- *Supabase or MongoDB Atlas database is configured*
- *GitHub repository is available*
- *Code is properly organized*
- *Project runs without errors*
- *Student is ready for viva*

Viva Preparation Questions

- *What problem does your project solve?*
- *Which database did you use and why?*
- *How many API endpoints did you create?*
- *Which APIs perform CRUD operations?*
- *How did you connect FastAPI with the database?*
- *Where did you deploy your backend?*
- *Where did you host your database?*
- *What challenges did you face during development?*
- *What improvements can be added in the future?*